Does Inflation Targeting Help Information Transmission?*

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Abstract

This paper studies the informational impact of inflation targeting (IT) on financial market activity and volatility in an emerging market context by using a unique, natural policy experiment. We evaluate the impact of a hard switch to IT by India in 2015, on the information transmitted to markets by central bank policy announcements. We find that central bank monetary policy exerts a significant impact on bond market trading and yield volatility, largely reflecting new information provided by policy decisions, but without a significant change in the informational properties of monetary policy surprises after the switch to IT barring a modest increase in wait-and-watch behavior before monetary policy decision dates. Monetary policy announcements and surprises do not significantly impact equity return volatility before and after the switch to IT. Our results point to market perceptions of monetary policy being informative about future inflation, but not the future growth outlook and risks. To support this intuition, we provide evidence based on textual analysis of Reserve Bank of India (RBI) policy announcements, which shows increase in focus on inflation but not growth.

JEL Codes: E44, E58

Keywords: inflation targeting, monetary policy, information transmission

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1 Introduction

Monetary policy decisions exert significant real and financial impact. Whether they be acts of commission; i.e., policy changes, or acts of omission; i.e., status quo maintenance, such decisions transmit information about the economic outlook, potential risks and the future direction of policy that is held by the policymakers to the market. Unless fully expected, policy announcements alter the borrowing costs for financial institutions and lead to a change in financing conditions for other economic agents. Anticipating this, financial markets respond immediately after policy announcements, which gets reflected in changes in trading volumes, asset returns and market volatility. Cross-country evidence (Bernanke & Gertler (2000), Jarocinski & Karadi (2018)) supports the view that policy announcements, such as a change in the policy interest rate or in the stance of monetary policy, impact both returns and volatility in the bond and stock markets.

An implicit assumption in the preceding paragraph is that monetary policy reacts credibly and predictably in response to information about the future evolution of the economy. It allows the market to invert the signal provided by the policy decision, infer the underlying information and respond with portfolio adjustments, which then trigger changes to asset returns and volatility. An implication of this chain of actions is that if the monetary policy framework changes to one that serves to enhance credibility and predictability, thereby changing the timing and nature of information transmission via policy decisions, then such a change in framework will impact financial market trading, returns and volatility.

Our paper uses an event study framework to understand the impact of IT on central bank information transmission. We examine the switch to flexible inflation targeting (FIT) by the Reserve Bank of India (RBI) in March 2015 to evaluate the existence of a differential impact on market outcomes of information released by policy decisions under the FIT regime versus the pre-FIT regime. We test for whether the IT framework which, in theory, should clarify the central bank's policy objectives and improve its communication, has helped the market in better understanding signals from the central bank. The Indian case provides, in our view, a compelling example since it was a hard change in the information release policy. Before FIT, the market did not possess precise information about the RBIâs monetary policy target, nor was there a consistent method of releasing information through monetary policy.

Our empirical exercise evaluates the impact of the changeover to FIT on trading volumes in the bond market and on the volatility of returns in Indian bond and stock markets. We use 10-year benchmark Government of India bonds, the most liquid segment of the Indian bond market and the large-cap, NIFTY-50 equity index for our analysis. We model the volatility in these markets as a GARCH process with a deterministic scale factor.

We find clear evidence that monetary policy has exerted a substantial effect on bond market activity in India throughout the last decade, both before and during the FIT regime. The volatility of bond market returns decreases immediately prior to the policy announcement and increases immediately after the announcement if the policy decision was not fully anticipated.[1] However, the central result of our analysis is the finding that the switch to FIT increased neither, the response of bond market trading or yield volatility over and above the impact that policy announcements exerted under the pre-FIT regime.

Turning to equity markets, neither in the pre-FIT period, nor after the switch to FIT do we see a systematic and visible impact on equity return volatility of monetary policy announcements and surprises. We find no evidence of a significant policy day effect in terms of higher volatility of stock returns of large-cap Indian firms. As with bond markets, there is no evidence supporting the case for an increase in information transmitted by RBI monetary policy after the switch to FIT that is perceived by markets as being significantly relevant to the pricing of large cap equity.

A possible explanation for the absence of a significant increase in response of Indian

[1] Our definition of the information content of monetary policy decisions accounts for the fact that some decisions may, in full or in part, be anticipated by markets. For example, monetary policy may respond to information contained in asset prices about future developments in inflation and output (Bernanke & Gertler (2000)). It may also reflect anticipated developments in macro-financial variables, including asset returns and volatility (Romer & Romer (2004)). Hence, the information content of monetary policy is not the change in the policy rate and stance, but instead, an induced change in a market price since this induced change will reflect the incorporation of new information.

financial markets to information released by monetary policy during the FIT regime could be that significant financial frictions are preventing prompt and efficient information absorption or action in response to such information absorption. However, this argument appears less salient in the context of large cap equity markets in India which are known to have significant depth and liquid trading conditions (Naik *et al.* (2020)). Moreover, when we test for the impact of U.S. monetary policy surprises on Indian markets, we find evidence of significant policy day effects on equity market volatility. This appears to rule out financial frictions as a significant factor behind the lack of response in equity markets to monetary policy surprises during pre-FIT and FIT regimes.[2]

A second possible reason for the lack of increase in the equity market's response to policy surprises during the FIT regime is that monetary policy is not significantly informative about the growth outlook and risks. This may be due to a combination of structural and cyclical factors. It could reflect the preeminence of fiscal policy in driving economic activity in emerging markets and developing countries (EMDCs) like India which would translate into a modest or negligible impact of monetary policy in affecting growth which inhibits the information content of monetary policy (in comparison with fiscal policy) from the perspective of valuation of large corporate firms.[3] From a cyclical standpoint, the FIT regime in India coincides in its entirety with a period of significant dislocation in credit markets due to the financial challenges confronting the banking sector and financial and operational challenges in the infrastructure and power sectors. Bank credit supply and investment activity by the infrastructure sector are key growth drivers in India, and in the FIT period, both have been under pressure.

We also test whether our result showing the absence of increased market reaction to

[2]These findings are robust to many alternative specifications, including limiting attention to monetary policy announcements involving repo rate changes, to studying the effect of policy decisions on foreign portfolio investment flows and other specifications.

[3]The long-term comovement of equity returns and economic growth is well-grounded from both theoretical and empirical perspectives. As argued by Bernanke & Kuttner (2005), a statistically significant response of equity returns to monetary policy surprises likely reflect the market-inferred changes in growth expectations.

monetary policy during the FIT period may be associated with a less-than-material change in RBI communication. To test this, we perform a textual analysis of RBI policy statements. We measure if the emphasis on growth and inflation changed after the switch to FIT. We use the text corpora from FIT and pre-FIT periods to create bi-grams (Picault & Renault (2017)). Next, we classify them into three groups, growth-related, inflation-related, or neither. We test if the frequency of growth-related and inflation-related bi-grams saw a change between the two periods as a measure of importance of these topics in RBI communication. Our analysis suggests no significant difference in the emphasis on bi-grams related to growth; however, the RBI emphasis on inflation witnessed an increase. This provides an additional explanatory channel supporting our interpretation of the results that the switch to FIT did not materially impact the market perceptions of policy information transmission related to growth. That markets do not appear to derive more growth-relevant information from monetary policy announcements in the FIT regime may reflect the fact that, in the context of fiscal preeminence, the central bankâs communication regarding growth and monetary policy did not materially change despite the growth outlook and risks being important part of the RBI's statements around the decision dates.

Overall, our paper makes two significant contributions to the literature. First, we study how market behavior around policy announcements changes after a hard switch to an IT policy regime. Our paper is unique since it utilizes a hard switch to IT regime in India. The properties of the pre-FIT regime give us a better benchmark against which to compare the impact of information release under FIT. Our paper joins the rich literature on the information content of monetary policy decisions and its impact on financial markets and the real economy. For the U.S., our paper is closest to Bomfim (2003), Bernanke & Kuttner (2005), and Gürkaynak *et al.* (2005), who assess the domestic stock marketâs response to monetary policy announcements. Similarly, Roley *et al.* (1998) evaluate the reaction of the stock market to monetary policy non-announcements. Most of these papers show how the news effect on stock market returns and volatility depends on a variety of factors, including the state of the business cycle, the direction of monetary policy shocks, and whether there is a bull or bear run in the market. However, insofar as we are aware, we are the first to directly study the role that inflation targeting can play in increasing the responsiveness of markets to news contained in monetary policy decisions and the resulting implications for the dynamics of trading and volatility around the decision date.

Second, our study extends the literature on the market response to information released by monetary policy in advanced economies to the case of EMDCs. In emerging economies, monetary policy signals may be noisier and financial markets have lesser depth. Hence, it is unclear ex-ante whether a change in the monetary policy framework which serves to clarify policy objectives and the policy reaction function would exert in EDMCs, the effects which are predicted by theory and supported empirically for advanced economies. The use of the regime switch by the RBI helps us in cleanly identifying the impact of a change in information release policy in a major emerging market economy which, given that IT is increasingly adopted by EMDC central banks, fills an important gap in understanding its efficacy for them. Outside the U.S. domestic context, Faust *et al.* (2003) propose an empirical framework to identify the response of non-U.S. output, interest rates, and the exchange rates to U.S. monetary policy surprises. Zare *et al.* (2013) assess the impact of monetary policy shocks on financial markets in Asian emerging market countries. Previous papers on EMDC markets (Kim *et al.* (2014) and Sun (2020)) do not use a regime switching empirical strategy nor study the impact of inflation targeting. Thus, our paper is the first one to examine whether IT improves market response to the information contained in monetary policy decisions in EMDCs.

Our paper is also related to recent work on central bank communication and its impact on the expectations and actions of various agents in the economy. The recent papers by D'Acunto *et al.* (2019) and Coibion *et al.* (2018) evaluate how much of this information from the central banks is actually absorbed by these agents. For example, Coibion *et al.* (2018) uses randomized controlled trial to show how firms in Italy with better information about central bank policy react more in line with the theoretical predictions of the models with rational agents. Our paper studies whether an IT regime can help improve central bank communication in the case of EDMCs. We present the institutional details about the RBI and the switch to FIT in the next section. In Section 3 and 4, we discuss our datasets and empirical framework, while in Section 5, we give our main results. Section 6 concludes.

2 Background

In this part, we describe the monetary policy framework and the changes instituted by the RBI after adopting FIT. We also conduct textual analysis on RBI's monetary policy statements to gauge if the central bank communication followed suit with these institutional changes.

2.1 Monetary Policy Framework: Details and Changes

Our paper studies the market reaction to decisions taken by the RBI and how it varies across the two periods, pre-FIT and FIT. The policy decisions are defined to include the choices pertaining to the level of the policy interest rate as well as public announcements regarding the stance of monetary policy. These announcements contain a qualitative description of the evolution of future policy rates, given the central bank's explicitly stated view of the likely trajectory of the economy and risks surrounding the baseline outlook.

India formally adopted the FIT framework in 2015.[4] Following the signing of the Monetary Policy Framework Agreement (MPFA) between the Government of India and the RBI in February 2015, the nominal anchor and operating target of monetary policy changed to

[4]The Government of India and Reserve Bank of India signed the Monetary Policy Framework Agreement on February 20, 2015 and fixed the target inflation rate of 4 percent with an error band of +/- 2 percent (https://www.finmin.nic.in/sites/default/files/MPFAgreement28022015.pdf). The first Monetary Policy Committee was constituted on August 5, 2016 (http://pib.nic.in/newsite/PrintRelease.aspx?relid=148405) after an amendment to the RBI Act, 1934 by Finance Act, 2016. We consider the former date as the start of the inflation targeting regime as the formal inflation target was fixed and RBI was given the mandate to achieve it.

maintaining the headline CPI inflation at 4 percent for the financial years 2016-17 through 2020-2021 with a band of +/- 2 percent. Practically speaking, given external pressures in the wake of the taper tantrum, the RBI had already established a formal framework in 2014 to guide monetary policy operations. It included a disinflationary glide path to below-6-percent CPI headline inflation by January 2016, in line with the FIT operational target.

Between May 2011 and February 2015, a period we call the Pre-Inflation Targeting (Pre-FIT), the operational target of monetary policy was the weighted average overnight call money rate (WACMR). The repo rate was in the middle of a 200-basis-points corridor, flanked by the reverse repo rate at the bottom and the marginal standing facility rate at the top. Finally, the liquidity management framework was revised to strengthen monetary transmission by anchoring the WACMR close to the policy repo rate. However, the repo rate constituted the main instrument of monetary policy in both, the pre-FIT period and the FIT periods.

The approach to communicating monetary policy decisions has also undergone major changes. The timing and content of policy announcements have changed since the adoption of FIT. During most of the pre-FIT period, there were two types of reviews of monetary policy, Quarterly and Mid-Quarterly. In the Quarterly reviews, which generally took place during the first week of a quarter, the policy meetings took place over two days, with the RBI releasing a written statement on the state of the economy a day before the final announcement on the policy stance. The statement on the state of the economy contained the central bank's views regarding various downside/upside risks to the baseline outlook for growth, inflation, exchange rate, etc., which in combination, provided justification for the policy stance announced on the next day. Hence, in the pre-FIT period, during the Quarterly reviews of monetary policy, the market already obtained important clues about the policy to be announced the next day. By contrast, the Mid-Quarter review policy meetings took place on a single day with a statement about the policy stance and, compared with the statement released during the Quarterly reviews, a short justification for the stance released simultaneously with the policy decision. In order to account for this different information release policy during the pre-FIT period in our empirical analysis, we control for the dates immediately around the two-day information release period (i.e., the dates when the statement on economic outlook and risks and the policy stance were respectively announced) separately in our regression specifications.

Another important difference between the pre-FIT and FIT periods lay in the absence of a pre-announced calendar for upcoming monetary policy meetings during the year for a majority of the duration of the former policy framework. Instead, the date of the next policy meeting was announced alongside each Monetary Policy Statement. This practice was given up in April 2014, since when Monetary Policy Reviews started to be held bi-monthly and policy stance along with justification for it being announced simultaneously on the same day. The practice of announcing the next policy date along with the Monetary Policy Statement continued, however, and the regime moved to a fully pre-announced calendar only after the formation of the Monetary Policy Committee in August 2016.

Overall, the switch to FIT entails three significant changes. First, the inflation target of the RBI is now explicit public information. Second, policy decisions are taken only on a single, pre-announced day. Third, it led to clearer communication since the RBI now must explain each of its policy stances to the public in a standardized manner.

By way of suggestive evidence for improved information transmission, we find that the switch to FIT is associated with the firming up of market expectations regarding the likely evolution of interest rates and is associated with lower average forecast errors regarding policy decisions. Comparing statistics from the RBI's quarterly survey of professional forecasts to actual policy-induced repo rate changes, we find that the average (mean and median) forecast errors fell significantly during the FIT period relative to the pre-FIT period (Panels (a) and (b) of Figure 1). Moreover, the uncertainty in market forecasts of the repo rate, measured as the difference between the market's maximum and minimum forecast rate, also fell in the FIT regime relative to the pre-FIT period, as seen in Panel (d) of Figure 1.

2.2 Did FIT Lead to a Change in Communication by the RBI?

The improvement in professional forecasters' performance suggests some changes in communication by the RBI during the FIT period. However, it does not provide an answer to the nature of this change. Therefore, we conduct a textual analysis of monetary policy statements issued by the RBI between 2011-2019. We analyze how the switch impacted RBI's focus on two main policy variables, growth, and inflation.

2.2.1 Measuring Importance of Inflation and Growth

We collect all the monetary policy statements from the RBI released at the time of monetary policy announcements between 2011-2019 (Table 2). These statements are shorter relative to the minutes of the MPC meetings or quarterly review published by the RBI. We focus on them because they are the main source of information on which the market reacts immediately after the policy announcement. The minutes of the MPC meetings and quarterly review are not released on the day of the policy announcement.[5]

We compute the frequency of inflation- and growth-related phrases to gauge the importance of these two themes in the policy statements. We are primarily interested in documenting whether these themes witnessed any change after the adoption of FIT. Unlike the recent work by Hansen *et al.* (2018) and Gentzkow & Shapiro (2010), we do not use advanced statistical techniques for theme detection, as our goal is to study the change in focus of RBI statements on growth and inflation after FIT, rather than the overall theme/sentiment classification of policy statements.[6]

Lexicon Building: We manually classified each one- and two-word phrase into growth and inflation categories with the RBI policy statements. Our approach is similar to Picault & Renault (2017), who classify phrases in ECB statements into monetary policy and economic outlook categories. However, they go further and also assign sentiment (hawkish, neutral

[5]The minutes of the meeting are released only in the FIT period. The minutes are published two or three weeks after the policy announcement.

[6]LDM or LSV can be used for un-supervised theme detection within a text.

and dovish) to these categories, which requires some subjectivity. Our exercise only requires us to calculate the relative weight that RBI policy statements give to growth versus inflation.

Before classification of phrases, we follow the standard text processing protocol to convert raw text into usable format. We convert all words to lower case, use the porter algorithm for stemming and removing all stop words. Finally, we are left with a total of 10,391 bi-grams in the whole period with 1023 of them unique (see table **??**). We then compute the total number of times each phrase appears in the entire text (both in the pre-FIT and FIT).

For classification, we restrict our attention to bi-grams in the entire corpus. The list of phrases classified under the two categories are given in Table **??**. For instance, CPI inflation, consumer prices, and the crude price are classified under inflation related phrases, and credit growth, external sector, and aggregate demand are classified under growth related phrases. The full list of bi-grams in provided in the Appendix Table A.4. We find that there is a common set of words which appear for both categories in both the BFT and FIT periods, but there is also a slight change in the vocabulary over time.

2.2.2 Inflation (not Growth) Became More Salient in FIT

To formally test the impact of FIT on growth and inflation related phrases in RBI statements, we calculate the Pearson's χ^2 statistic given by:

$$\chi^2 = \frac{N(ad - bc)^2}{(a + c)(b + d)(a + b)(c + d)} \tag{1}$$

where a, b, c and d are as defined in the contingency Table 8. The above χ^2 statistic has 1 degree of freedom and is used to compare the frequency of words across two corpus. For example, Gentzkow & Shapiro (2010) use this statistic to classify phrases that are more commonly used by Republican or Democratic politicians in their speech. One crucial difference and innovation in our approach is that we do not use the above test to differentiate whether the individual word *growth* appears differently in the two corpora. We instead pool together all the bi-grams related to growth and then test for the difference in their appearance in the text. We compute the χ^2 statistic for both growth and inflation related phrases.

The frequency distribution of keywords over the two periods is given in Table 9. In Panel (a) we provide the contingency table for growth related phrases. We find no significant difference in the growth related keywords in monetary policy announcements between the two periods. The frequency of growth related bi-grams increases slightly during the FIT period, but the χ^2 test shows that it is not a significant change. In contrast, there is a significant increase in the keywords related to inflation (Panel (b)). The χ^2 test has a value 12.8 and is significant. The frequency of inflation related bi-grams jump from 15.2 percent in the BFT to 18.0 percent in the FIT period.

It is also important to mention that realized inflation is much lower during the FIT period relative to the pre-FIT period. However, the RBI statements still put more emphasis on inflation in the FIT period, as captured by the inflation related bi-grams. This change perfectly aligns with the goal of an inflation targeting regime. At the same time, there was no change in the focus on growth. The keywords associated with growth are more than twice as frequent as those associated with inflation in both periods. However, in terms of overall focus on growth, policy statements did not change in FIT.

Overall, these results provide evidence that the switch to the FIT regime was followed up by actual changes in communication by the RBI. However, the communication focus has relatively increased only in the case of inflation and not growth. How the markets perceived these changes is something we test in the following sections. We begin by describing our primary datasets.

3 Data

We use multiple sources of data in our empirical exercise. The first set consists of yield and returns data. We use daily data on benchmark G-Sec yield for the 10-year benchmark government bond and NIFTY-50 index from the National Stock Exchange between January

2011 and August 2019. The G-Sec yield data is taken from the Clearing Corporation of India Limited while the NIFTY data comes from the National Stock Exchange. Finally, our measure of policy surprise is constructed using the one-month ahead over-night interest rate swap (OIS). The OIS data comes from Bloomberg. Finally, we use the repo rate information from the RBI.

Next, we hand-collect the dates of monetary policy announcements for the same period from the Reserve Bank of India website. Our final dataset consists of a total of 2072 business days, of which 999 are from the BFT period while 1073 are from the FIT period. There are a total of 60 policy days during this period, out of which 31 days belong to pre-FIT period and the rest to the FIT. There are 26 policy days with a change in the repo rate, with 16 belonging to the pre-FIT and 10 to the FIT period.

We also use additional datasets related to the FOMC and FPI flows in India. We collect information on FOMC meeting dates and Fed Funds Future rate for January, 2011 - August, 2019, from Bloomberg. The Foreign Portfolio Investment data on debt and equity for India comes from Bloomberg.

Sample Stratification: As discussed in the introduction, the switch to the FIT framework entailed a change to the nominal anchor/target of monetary policy and corresponding operational changes to the conduct of monetary policy. Consequently, we stratify the longitudinal sample into two periods, one corresponding to the pre-FIT period (January, 2011 - February, 2015) and the other corresponding to the FIT period (March, 2015 - August, 2019). Moreover, within the FIT period, the months between November 2016 to June 2017 (all inclusive) need a separate assessment since they bore witness to the major exogenous shock of demonetization which may have exerted a potentially confounding influence on the information content of RBI monetary policy decisions and on the relationship between informativeness and market response in terms of trading and asset price volatility. We drop the 3 policy days corresponding to the demonetization period.

4 Empirical Framework

The main objective of the empirical exercise is to test whether the days around the monetary policy decision of the RBI display heightened volatility in the bond and stock markets. Since we study the impact of FIT on both these markets, it is essential to highlight the context behind this choice.

Bonds vs. stocks: In the advanced economies, the success of IT has meant a significantly greater analytical interest in the impact of monetary policy surprises on the price of risky assets rather than benchmark (government) bonds. The monetary policy in these countries is informative about the evolving growth outlook, it should be expected to have a predictable impact on equity returns, and hence, it should trigger an immediate and short-lived increase in stock trading and market return volatility. Developing countries are a different kettle-of-fish. While domestic monetary policy does exert influence on local financial conditions, policy transmission is imperfect and incomplete (IMF, 2017), even in the bond market. The constraints on transmission of domestic monetary policy in emerging market economies relative to the advanced economies could be an important contributer to well-known differences in business cycle properties across the two groups of countries (Aguiar & Gopinath (2007)).[7] Indeed, as has been recently argued, emerging market countries may be confronted with a policy dilemma, wherein, even with floating exchange rates, monetary policy may have limited room for maneuver in the presence of free and volatile global capital flows that are heavily influenced by a common global factor linked to U.S. monetary policy (Obstfeld (2015);Rey (2015)).

Consequently, we first test the impact of monetary policy announcements on the bond markets and subsequently extend the analysis to the stock market.

[7]Equities and (government) bond markets are mature, deep and liquid in advanced economies. For example, while US monetary policy shocks impacts price of risk globally (Miranda-Agrippino & Rey (2015)), monetary policy in developed countries, including small open economies, is able to largely control domestic financial conditions due to a combination of local financial market depth and floating exchange rates.

4.1 Impact of Policy Announcements on Trading of Government Securities

Before we analyze the impact of FIT on volatility, we first do a preliminary test on the total volume of government securities traded in the bond markets. If the adoption of FIT has improved information transmission, we should expect market activity to be at a lower level immediately prior to the decision date, and at a higher level on the policy decision date during the FIT period but not during the BFT period. We use the following event study regression framework to test it:

$$\begin{aligned} Growth(Volume)_t = \alpha_0 + \alpha_1 I_t^{MPC-} + \alpha_2 I_t^{MPC} + \alpha_3 I_t^{MPC+} + \alpha_1^{'} I_t^{MP2--} \\ + \alpha_2^{'} I_t^{MP2-} + \alpha_3^{'} I_t^{MP2} + Week_{FE} + \varepsilon_t \end{aligned} \quad (2)$$

where $Growth(Volume)_t$ is the dependent variable and is equal to the growth in daily volume traded in the government securities market. We consider trading in the benchmark 10-year sovereign bonds as it is the most liquid market. This growth depends on three variables I_t^{MPC-}, I_t^{MPC} and I_t^{MPC+}. I_t^{MPC-} is a dummy variable that equals one if t falls one day before the MPC, else it is zero. Similarly, the dummy variables I_t^{MPC} and I_t^{MPC+} take a value equal to one on the day of MPC meeting and one day after the MPC meeting respectively, and zero otherwise. We also include additional dummy variables for the policy days when policy was declared over two days through the variables I_t^{MP2}, I_t^{MP2-} and I_t^{MP2--}. Here I_t^{MP2} is equal to 1 on the final policy day, while I_t^{MP2-} is equal to 1 on the day pre-policy statement was released by RBI and I_t^{MP2--} is equal to 1 on the day prior to that. On all other days, these variables take a value zero.

If policy days are special and actually drive higher trading volume on the market, then the coefficient α_2 should be positive. If the market behaves like any other trading day on the day prior to the policy meeting, then α_1 should be zero. If instead trading is subdued on this day then α_1 will be negative. Finally, if the policy day has a higher trading activity, i.e. α_2 is positive, then the next day can have a negative volume growth even if the trading volume is average on that day. Due to this reversal in level effect, α_3 can be negative purely because of a positive α_2. Since our goal is to test whether the trading behavior changed during the FIT period, we include dummy variables for I_t^{MPC-}, I_t^{MPC} and I_t^{MPC+} interacted with the FIT dummy. We expect the coefficient on $FIT \times I_t^{MPC}$ to be significantly positive for the FIT period if the market associates RBI policy releases to have superior information content during this period.

Finally, the fixed-effect term $Week_{FE}$ captures the impact of unanticipated economic and policy developments other than monetary policy, but which impinge upon agents' portfolio allocation and liquidity management decisions.

4.2 Impact of Policy Announcement on Volatility

In this sub-section, we describe our framework to study the impact of policy days on market volatility. We use various models to estimate the impact of policy announcements on market volatility.

Model 1: The baseline empirical model is given by the following set of equations:

$$r_t = \alpha + \beta x_t + u_t \tag{3}$$

$$E\left[u_t^2|\Omega_{t-1}\right] = log(s_t) + log(\sigma_{t-1}^2) \tag{4}$$

$$s_t = exp[\delta_1 I_t^{MPC-} + \delta_2 I_t^{MPC} + \delta_3 I_t^{MPC+} + \mu_1 I_t^{MP2--} + \mu_2 I_t^{MP2-} + \mu_3 I_t^{MP2}] \tag{5}$$

where the dependent variable r_t in equation 3 captures either the evolution of bond yields or stock market returns. This r_t depends on a vector of variables given by x_t that includes a lagged value of the dependent variable and a measure of future expectations about the monetary polciy. This specification is similar to Bomfim (2003) with one major difference i.e. we use the Overnight Interest Swap (OIS) rate of one month maturity as a measure of future expectations about monetary policy. This is necessitated by the fact that unlike the US, India does not have an active, representative futures or forward market. The unpredictable component of r_t; i.e., u_t, has two components: (i) a stochastic component, σ_t^2, which, following Nelson (1991), is assumed to follow an EGARCH (1,1) process with a multiplicative heteroskedastic term and (ii) a deterministic scale factor, s_t, which provides the main channel for policy announcements to have a separate effect on the volatility.[8] The deterministic scale factor, s_t, in equation 5 depends on three variables, I_t^{MPC-}, I_t^{MPC} and I_t^{MPC+}. These variables are defined in the same way as in equation 2. Consequently, the normal level of the scaling factor for bond yield or stock return volatility is unity which is allowed to vary around the policy dates. To capture differential effect during the two day policy regime, we have also included dummy variables, I_t^{MP2}, I_t^{MP2-} and I_t^{MP2--}, in equation 5.

The main coefficients of interest in *Model 1* are δ_1, δ_2 and δ_3, which together capture the impact of scheduled policy announcement days by RBI on volatility of r_t. If the policy days are indeed special and deliver information to the market, then δ_2 should be positive and I_t^{MPC} should be associated with a higher volatility. At the same time, if there is a *calm before the storm effect* then $\delta_1 < 0$. Finally, δ_3 should be zero as the market should incorporate all the relevant information on the policy day itself, and there should be no impact on volatility on the day after the policy.

Model 2: It is possible that the market is volatile on the day of the policy, but any expected information should not drive up this volatility. If the information released is ex-

[8] As in Bomfim (2003, p. 138), the model, combined with the definition of s_t implies that the conditional variance of u_t (and hence, bond yield or stock return volatility), will exceed the volatility of the simple EGARCH (1,1) model on days when the scaling factor exceeds unity. In particular, MPC dates which release information to the market and push up the value of s_t, would result in higher market volatility on the day.

pected, then it should already be accounted for by market participants prior to the policy announcement. Instead, market volatility should be more pronounced only on the days when the policy day is also associated with an informational surprise. To test this, we modify equation 5 to incorporate a surprise effect in the following way:

$$s_t = exp[\delta_1 I_t^{MPC-} + \delta_2 I_t^{MPC} + \delta_3 I_t^{MPC+} + \delta_4 I_t^{S} \\ + \mu_1 I_t^{MP2--} + \mu_2 I_t^{MP2-} + \mu_3 I_t^{MP2}] \quad (6)$$

Here, we capture the surprise effect through the variable I_t^S. It is a dummy variable which takes a value equal to one when the policy announcement day is associated with a surprise, otherwise it is zero. We describe how we calculate surprise in detail through OIS in the next section. In *Model 2*, the additional coefficient of interest is δ_4 which should be positive if the policy days with a surprise are associated with a heightened market activity as well as volatility.

4.3 Evaluating the change due to FIT

To analyze whether FIT led to any change in market reaction to policy surprises, we augment the models described in the previous sub-section. We first test if there is any additional impact due to the FIT itself.

Model 3: We modify equation 5 to capture the impact of FIT on market volatility in the following way:

$$s_t = exp[\delta_1 I_t^{MPC-} + \delta_2 I_t^{MPC} + \delta_3 I_t^{MPC+} + \mu_1 I_t^{MP2--} + \mu_2 I_t^{MP2-} + \mu_3 I_t^{MP2} \\ + I_t^{FIT} \left(\gamma_1 I_t^{MPC-} + \gamma_2 I_t^{MPC} + \gamma_3 I_t^{MPC+}\right)] \quad (7)$$

where I_t^{FIT} is a dummy variable equal to one for the FIT period. Since I_t^{FIT} is interacted with policy day dummies in equation 4.3, the additional impact on volatility due to FIT is captured by the coefficients γ_1, γ_2 and γ_3. If γ_2 is positive, it will highlight that the FIT period is associated with an increase in the volatility on the policy day, which is over and above what is captured by δ_2. Instead, if it is zero then the FIT period is no different from the BFT period on the policy days. Finally, we evaluate the impact of information in *Model 4* below.

Model 4: It builds upon *Model 3* and incorporates the additional effect of surprise during the FIT period. Once again to capture this effect, we modify equation 6 as below:

$$
\begin{aligned}
s_t = exp[\delta_1 I_t^{MPC-} + \delta_2 I_t^{MPC} + \delta_3 I_t^{MPC+} + \mu_1 I_t^{MP2--} + \mu_2 I_t^{MP2-} + \mu_3 I_t^{MP2} \\
+ \delta_4 I_t^S + I_t^{FIT} \left(\gamma_1 I_t^{MPC-} + \gamma_2 I_t^{MPC} + \gamma_3 I_t^{MPC+} + \gamma_4 I_t^S\right)]
\end{aligned}
\tag{8}
$$

where relative to *Model 3*, the additional coefficient of interest is γ_4. It captures whether the surprise has an additional impact on volatility in the FIT period, which is over and above the impact captured by δ_4. A positive sign on γ_4 will indicate an additional effect of surprise, while a negative sign will reflect vice-versa.

5 Empirical Results

In this section, we present the results from estimating the models presented in Section 4. We begin by describing how OIS can be used to measure the surprise component of monetary policy announcements in the Indian context.

5.1 Using OIS to measure Monetary Policy Surprise

The conventional approach of calculating the information content of policy announcements is to use the change in interest rate futures. However, this is not possible in the Indian context as there is no standard futures contract available for trading. Moreover, even in cases where futures are traded, it is usually done to support the liquidity management of banks' balance-sheets rather than extending to rates-trade arbitrage, short-sales, or trading in specials. Given these constraints, we use the change in the spread on the OIS contract as a measure of surprise information.[9] To construct the policy surprise dummy, I_t^S, we follow Kuttner (2001) and assign a value of one to days associated with the change in OIS above a certain threshold.[10]

Since the change in OIS is a non-standard way of measuring the information content of policy in the literature, we first provide evidence in favor of its validity. Figure 2 plots the time trend of one-month OIS spread which provides preliminary evidence in this regard. The vertical lines on the graph correspond to MPC policy days and the blue shaded area corresponds to a 30 day period before the contract ends. We provide a snapshot of the OIS spread from May-December 2019 when the market expected a rate cut from the RBI each time. As can be clearly seen from Figure 2, the evolution of OIS captures this sentiment very well as it falls in each of the blue shaded areas up until the next policy announcement date. Importantly, since it is a 1-month OIS, it starts falling exactly 30 days before the MPC meeting.

As a second test, we formally check whether the change in the OIS spread is a good predictor of the change in the 10 year sovereign bond yield on policy days. The information that the market has about the policy announcement should already be accounted by the OIS spread on the day prior to the policy meeting. Only unanticipated information will change

[9] Trading in OIS contracts in India is apparently limited primarily to foreign banks who hold a minority of the fraction of underlying assets. The market is tiered, in the sense that whereas non-U.S. banks clear OIS trades through a domestic central counter-party, U.S. banks that constitute over 30 percent of all trading volume, engage in bilateral OTC trades, necessitating different counter-party risk management and capital requirements. Finally, in what is unconventional from a cross-country perspective, OIS contracts in India tend to trade at a premium relative to the underlying. These factors indicate limits to the interpretability of movements in the OIS spread as a measure of information flow to markets. Nonetheless, and particularly in view of the supporting evidence provided in this section for its use and of the absence of alternative measures, we view the use of OIS change as an acceptable and practical option to measure monetary policy surprises.

[10] In our main results, $I_t^S = 1$ on the policy days when OIS change is above its 75^{th} or below 25^{th} percentile value evaluated over the whole policy day sample. However, the results are similar to other threshold levels as well.

the OIS spread on the policy announcement date which should then perfectly explain the change in sovereign bond yield after the policy announcement. The results of this test are shown in Table 3, where we regress change in the 10 year bond yield on the expected and unexpected movements in the policy rate, i.e. repo rate. In column (1), we find that the change in the bond yield is positively correlated with the change in the repo rate. However, once we decompose this change in the repo rate into its expected and unexpected components, we see that it is only correlated with the latter. This unexpected component is the difference in the OIS rate on the day of policy and one day prior. Both the graphical assessment and this formal test give us good evidence in favor of using the OIS spread to capture the unexpected component in policy announcements in the Indian context.

5.2 Bond Market Trading Activity did not change after FIT

We report the results from estimating equation 2 in Table 4. Column (1) reports the result for the impact of policy days on daily trading volume growth. We find a fall in trading volume growth one day before the policy announcement and an increase on the policy date ,i.e., the coefficient on *One Day Before Policy* is negative and on the *Day of Policy* is positive. This goes in line with the *calm before the storm* and *storm* hypotheses. However, there is no differential effect on trading volume growth in the FIT period around the policy days as all the coefficients on $FIT\times$policy days are insignificant. We report the results with the inclusion of dummies for two-day policy dates in column (2). This specification has similar results and there is no differential impact on trading growth variable during the FIT period.

We further stratify the policy days into those days where the policy rate, i.e. repo rate, changes versus others where there is no repo rate change. The results for the case when the repo rate changes are reported in columns (3) and (4). We find that trading volume growth falls one day prior to the repo rate change dates and increases on such policy dates. However, we now see that during the FIT period, there is a higher *calm before the storm* effect, since the coefficient on $FIT\times$One Day Before Policy is negative and significant. Once we include the dummies for two-day policy dates, this effect is visible only during the FIT period. The main takeaway from this analysis is that trading activity is significantly different around policy dates, irrespective of whether the policy entails a change in the repo rate or not and the additional impact of the switch to FIT is reflected only in the decrease in growth of traded volumes immediately prior to the policy announcement.

5.3 No Change in Bond Market Volatility

Next, we estimate equation 5 and report the results on bond market volatility in Table 5. In column (1) we report the results based on *Model 1*. We find that the policy days are associated with a significant and positive impact on bond market volatility as the coefficient δ_2, on the Day of Policy, is positive and significant. However, there is no *calm before the storm effect* as the coefficient on One Day Before Policy is insignificant. There is also some negative effect on the volatility one day after the policy meeting.

Next, we report the results of estimating *Model 3* in column (3). We find that the coefficient on the day of policy is still significant. However, there is no additional effect during the FIT period, the coefficient on the policy day during FIT, γ_2, is insignificant. If greater information was being conveyed during the FIT period, the volatility in the bond markets would have been relatively higher on policy days during the period after the switch to this framework and we would have found this coefficient to be positive and significant. Similarly, there is no additional *calm before the storm effect* in the FIT period as the coefficient γ_1 is also insignificant.

Finally, to evaluate the impact of the policy surprise on volatility, we estimate *Models 2* and *4* and report their results in columns (2) and (4) respectively. We find that the surprise element in policy has an impact on volatility in both cases. The coefficient δ_4 is positive and significant, which implies that volatility goes up whenever the RBI's policy announcements contain a surprise element. In the case of *Model 2*, the effect of the policy day on volatility, i.e. δ_2, still remains positive and significant. So, the surprise element in the policy announcement has a significant positive effect on bond yield volatility over and above that of the policy announcement itself. However, compared to *Model 1* in column (1), the coefficient δ_2 is lower in magnitude which reflects the fact that a significant portion of the *storm* effect in *Model 1* owes to the surprise component of the policy announcement.

The results capturing the differential impact of FIT period are reported in column (4). We find that there is no additional impact of policy surprises during the FIT period, above and beyond the one present during the pre-FIT period. It is captured through the insignificant coefficient, γ_4.

Overall, our results suggest that the FIT regime has not had a significant impact on the bond market volatility.

5.4 RBI Policy Surprises do not Impact Equity Market Volatility

We report the results on the response of the stock market to monetary policy announcements and surprises in Table 6. Unlike the bond market, we find an increase in the volatility of stock returns one day prior to the policy announcement (positive and significant coefficient δ_1 for all *Models 1-4*). However, there is no increase in equity return volatility on the policy date itself, nor a significant impact of the surprise component of policy decisions. The one similarity of these results to the corresponding ones for the bond market is a decrease in stock return volatility one day after the policy announcement.

While there is no evidence of a *calm before the storm* effect during the FIT period either, we find evidence of a significant decrease in large-cap equity return volatility the day prior to the policy announcement compared to the pre-FIT period. This is reflected in negative and significant coefficient γ_1 in columns (3) and (4). It is in line with the increase in predictability and comprehensibility of monetary policy information transmission after the switch to FIT. However, a clean interpretation is still difficult for two reasons. First, volatility doesn't actually decrease before the decision date, only that it declined relative to before the switch to FIT. Second, there is no evidence of a *storm* effect; i.e., of markets trading more in response to information transmitted through the policy decision.

These results are intriguing, especially when put side-by-side with those for the bond market. The Indian large cap equity market, the NIFTY-50, is a relatively deep and liquid market, which means that we expected results similar to those in more advanced economies, i.e., we expected a response of stock market volatility to monetary policy information transmission. It begs the question of whether it is some type of financial friction (present in an emerging market like India but not in advanced market economies) or the market's perception of the absence of information regarding growth in monetary policy which is behind the negative result. While we do not address this question directly in this paper, we conduct two additional empirical exercises to argue that it the second factor; i.e., market perception that monetary policy is not informative about growth that is the likely explanation.

5.5 US Monetary Policy Impacts Volatility in India

The US monetary policy decisions and surprises can be expected to exert an effect on returns and volatility of most risk assets, including emerging market equities and sovereign bonds. If such an impact exists and is significant for Indian financial markets, this would be indicative of sufficient depth and sensitivity of these markets. The first of our empirical exercises examines whether the US monetary policy decisions and surprises has an impact on volatility in Indian financial markets. We estimate Models 1 and 2 to evaluate the impact on market volatility due to the information released around the FOMC (Federal Open Market Committee) meeting days, the MPC counter-part in the US.[11]

The results for the estimation of *Model 1* and *2* are reported in Table 7. The first two columns correspond to the impact on the domestic 10 year sovereign bond market around the Federal Open Market Committee policy meetings and announcement days and

[11] Since Indian markets are closed at the time of FOMC announcements, we consider the next immediate market day in India as the policy announcement day. The one day before and one day after FOMC meeting is then defined relative to this policy announcement day. Finally, we calculate the surprise element in US Fed policy by taking the difference in the Fed Funds Future rate between the day of FOMC meeting and one day prior to it. This is based on US time. Equivalently, it is the difference in Fed Funds 1-year Future rate between $t-1$ and $t-2$ market day according to Indian Standard Time.

the subsequent two columns correspond to the impact on the NIFTY-50 large cap equities market.

In the bond market, the day of policy effect is important as the coefficient δ_2 is positive and significant in column (1). However, this effect disappears once we account for the surprise component in FOMC meetings in column (2). The impact on bond market volatility is mainly coming through the surprise element as in *Model 2* only the coefficient δ_4 remains positive and significant. Additionally, we see no decline in bond yield volatility on the day before the FOMC announcement. Thus, the behavior of bond market volatility around FOMC days is broadly similar to days around the RBI policy announcement. The only crucial difference is that the policy day in itself is important for RBI announcements even after controlling for the surprise component of policy. Under the FOMC case, everything is explained by the surprise component of policy and policy day alone is not important. This difference can arise due to the difference in timing of announcements by the two institutions. The bond market in India already knows about the FOMC decision before the start of the trading day and needs to incorporate only the surprise component of FOMC decision in its trading behavior.

The results for NIFTY-50 stocks are quite different as reported in columns (3) and (4). We find a significant and positive day of policy effect on equity return volatility (coefficient δ_2 is positive and significant) in both models. This day of policy effect does not vanish once we include the surprise component of policy, which itself is insignificant ($\delta_4 = 0$ in column (4)). Unlike the bond market, we however find that the day prior to FOMC announcement witnesses a significantly negative volatility ($\delta_1 < 0$) in equity markets. These results are also in stark contrast when compared to the effect on stock market volatility around RBI policy days. The reaction of Indian stock markets to FOMC announcements is more in line with the reaction of US stock markets to FOMC announcements than its reaction to RBI announcements.

Overall, the main takeaway from our assessment of the impact of the US monetary policy on Indian financial markets is that both bond and stock markets respond to new information, especially in the case of the equities market.[12] Hence, large cap equities in India retain substantial depth so as to absorb and reflect new information through changes to trading volumes, returns and volatilities. It seems far more plausible, therefore, that the negative results for equities relative to the RBI's policy announcements reflect the market perception of lack of information therein for economic growth rather than the inability to trade on the basis of this information even if judged material to portfolio choice.

6 Conclusion

This paper evaluates the impact of the hard switch to IT by India in 2015 on the information transmitted to the markets through central bank policy announcements. We find that the switch to IT does not correspond to better information transmission to the market. The monetary policy in India was already informative to the bond market, and IT does not improve upon it. The equity markets have also not shown much change in their response. A possible explanation is that, in the Indian context, the switch to inflation targeting has rendered monetary policy action to be more predictive about future inflation but not necessarily about the growth outlook and risks.

[12] We show in the Appendix how FPI flows play a key role in transmission.

References

Aguiar, Mark, & Gopinath, Gita. 2007. Emerging market business cycles: The cycle is the trend. *Journal of political Economy*, **115**(1), 69–102.

Bernanke, Ben, & Gertler, Mark. 2000. *Monetary policy and asset price volatility*. Tech. rept. National bureau of economic research.

Bernanke, Ben S, & Kuttner, Kenneth N. 2005. What explains the stock market's reaction to Federal Reserve policy? *The Journal of finance*, **60**(3), 1221–1257.

Bomfim, Antulio N. 2003. Pre-announcement effects, news effects, and volatility: Monetary policy and the stock market. *Journal of Banking & Finance*, **27**(1), 133–151.

Coibion, Olivier, Gorodnichenko, Yuriy, & Ropele, Tiziano. 2018. *Inflation expectations and firm decisions: New causal evidence*. Tech. rept.

D'Acunto, Francesco, Hoang, Daniel, Paloviita, Maritta, & Weber, Michael. 2019. Human frictions in the transmission of economic policy. *University of Chicago, Becker Friedman Institute for Economics Working Paper*.

Faust, Jon, Rogers, John H, Swanson, Eric, & Wright, Jonathan H. 2003. Identifying the effects of monetary policy shocks on exchange rates using high frequency data. *Journal of the European Economic association*, **1**(5), 1031–1057.

Gentzkow, Matthew, & Shapiro, Jesse M. 2010. What drives media slant? Evidence from US daily newspapers. *Econometrica*, **78**(1), 35–71.

Gürkaynak, Refet S, Sack, Brian, & Swansonc, Eric T. 2005. Do Actions Speak Louder Than Words? The Response of Asset Prices to Monetary Policy Actions and Statements. *International Journal of Central Banking*.

Hansen, Stephen, McMahon, Michael, & Prat, Andrea. 2018. Transparency and deliberation within the FOMC: a computational linguistics approach. *The Quarterly Journal of Economics*, **133**(2), 801–870.

Jarocinski, Marek, & Karadi, Peter. 2018. Deconstructing monetary policy surprises: the role of information shocks. *ECB working paper*.

Kim, Hyeyoen, Kim, Junyeup, Lee, Jaeram, & Ryu, Doojin. 2014. The impact of monetary policy on banking and finance stock prices in China. *Applied Economics Letters*, **21**(18), 1257–1261.

Kuttner, Kenneth N. 2001. Monetary policy surprises and interest rates: Evidence from the Fed funds futures market. *Journal of monetary economics*, **47**(3), 523–544.

Miranda-Agrippino, Silvia, & Rey, Hélene. 2015. *US monetary policy and the global financial cycle*. Tech. rept. National Bureau of Economic Research.

Naik, Priyanka, Poornima, BG, & Reddy, YV. 2020. Measuring liquidity in Indian stock market: A dimensional perspective. *PloS one*, **15**(9), e0238718.

Obstfeld, Maurice. 2015. Trilemmas and trade-offs: living with financial globalisation. *BIS working paper*.

Picault, Matthieu, & Renault, Thomas. 2017. Words are not all created equal: A new measure of ECB communication. *Journal of International Money and Finance*, **79**, 136–156.

Rey, Hélène. 2015. *Dilemma not trilemma: the global financial cycle and monetary policy independence*. Tech. rept. National Bureau of Economic Research.

Roley, V Vance, Sellon, Gordon H, *et al.* 1998. *Market reaction to monetary policy nonannouncements*. Tech. rept. Federal Reserve Bank of Kansas City.

Romer, Christina D, & Romer, David H. 2004. A new measure of monetary shocks: Derivation and implications. *American Economic Review*, **94**(4), 1055–1084.

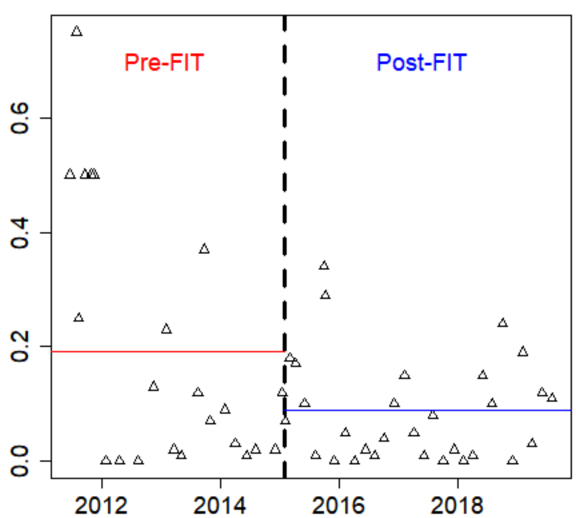
Sun, Rongrong. 2020. Monetary policy announcements and market interest ratesâ response: Evidence from China. *Journal of Banking & Finance*, **113**, 105766.

Zare, Roohollah, Azali, M, & Habibullah, Muzafar Shan. 2013. Monetary policy and stock market volatility in the ASEAN5: Asymmetries over Bull and Bear markets. *Procedia Economics and Finance*, **7**, 18–27.

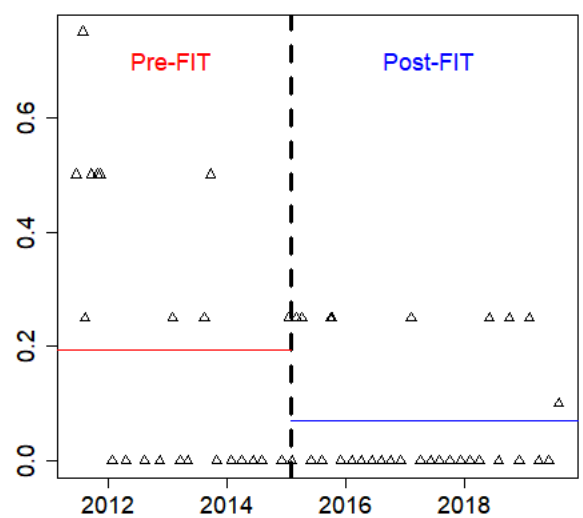
Figures

Figure 1: Performance of Professional Forecasters

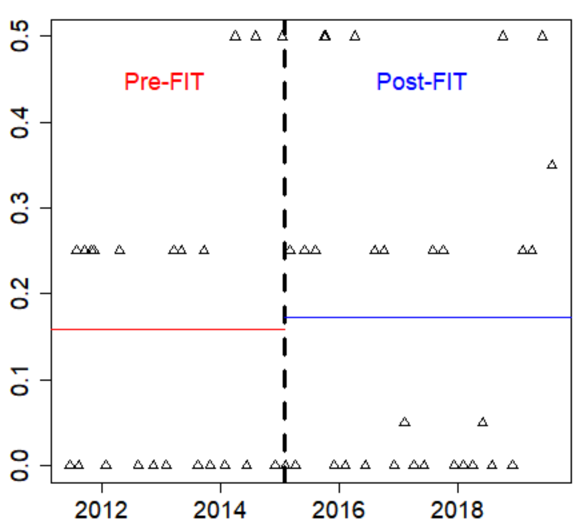
(a) Absolute (Error in Mean Forecast)



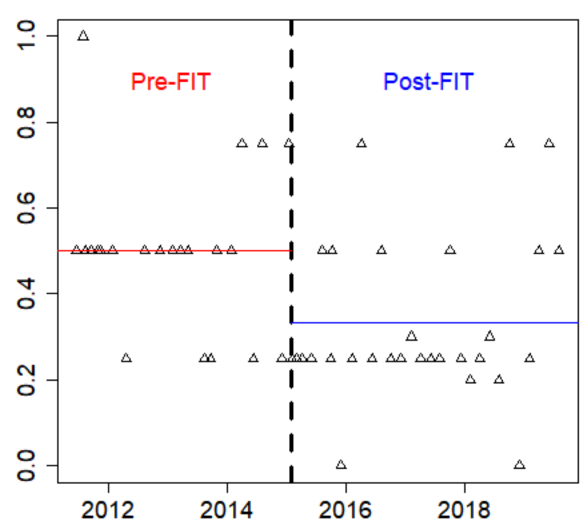
(b) Absolute (Error in Median Forecast)



(c) Absolute (Error in Max Forecast)

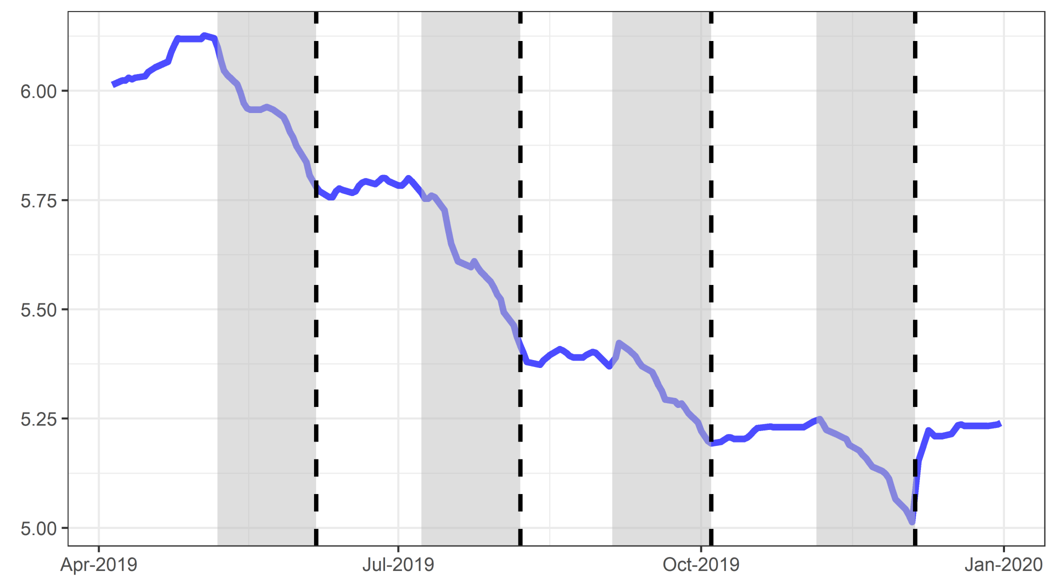


(d) Uncertainty (Max Forecast - Min Forecast)



Notes: Panel (a)-(c): Each Δ corresponds to the absolute value of the error between the given forecast moment and announced repo rate. The red line corresponds to the average value of the statistic in the Pre-FIT period and blue line corresponds to the average value in post-IT. Panel (d) gives the difference between the maximum and minimum value of repo rate forecasts. All data comes from the RBI Survey of Professional Forecasters.

Figure 2: 1-month OIS Provides Information About Monetary Policy Direction



Notes: The blue line corresponds to the 3-day moving average of 1-month OIS. The dashed vertical lines correspond to the RBI policy dates, while shaded grey corresponds to one-month periods before the policy dates.

Tables

Table 1: Descriptive Statistics on Pre- and Post-FIT Monetary Policy Meetings

	Pre-FIT	Post-FIT
	(1)	(2)
Meetings	32	28
Meetings with Repo Change	16	10

Notes: This tables gives summary on the number of Monetary Policy Meetings under the two regimes.

Table 2: Descriptive Statistics on Text of RBI Policy Statements

	Pre-FIT	Post-FIT
# of documents	32	28
# of words	47946	76772
Average length	1498.31	2741.86

Notes: The table provides summary information on the statements released by the RBI immediately after the announcement of its monetary policy.

Table 3: Surprise Component of Policy Impacts 10-year G-Sec Yield

	Log Change in G-Sec	
	(1)	(2)
1. β_0: Log Change in Repo Rate	-0.0040	
	(0.0362)	
2. β_1: Log Change in OIS (Unexpected Component)		0.2037***
		(0.0584)
1 - 2 β_2: (Expected Component)		-0.0208
		(0.0221)
N	57	57

Notes: The above table provides impact of repo and surprise repo change on 10-year benchmark G-Sec yield. The equations are estimated using 57 policy days (pre-announced ones) between 2011 and 2019. We use the following equation:

$$\Delta Log(Yield_t) = \alpha + \beta_0 \Delta Log(Repo_t)$$

$$\Delta Log(Yield_t) = \alpha + \beta_1 \Delta Log(OIS_t) + \beta_2 [\Delta Log(Repo_t) - \Delta Log(OIS_t)] \epsilon_t$$

Robust standard errors of the coefficients are reported in the parentheses. ***- $p < 0.01$, **- $p < 0.05$ and * - $p < 0.1$.

Table 4: No Additional Effect on 10-year G-Sec Volume around RBI Policy Days (after FIT)

	Growth in G-Sec Volume			
	All Days		Repo Change Days	
	(1)	(2)	(3)	(4)
One Day Before Policy	-25.5493***	-23.9035***		-22.1581*
-18.3776				
	(7.0181)	(7.0048)	(11.2986)	(12.8454)
Day of Policy	79.4230***	69.3041***	74.0667***	57.2484***
	(13.2699)	(14.3823)	(15.8353)	(17.3651)
One Day After Policy	-1.9486	-7.4753	-12.5359	-15.5175
	(8.0003)	(8.2846)	(9.7205)	(10.0107)
FIT× One Day Before Policy	-11.2635	-12.8549	-35.8523*	-39.6328*
	(15.2958)	(15.8429)	(19.7805)	(20.7412)
FIT×Day of Policy	1.5095	11.5734	0.8548	17.6731
	(23.2285)	(25.2062)	(35.2089)	(36.0055)
FIT×One Day After Policy	-10.1639	-4.6391	-36.9036	-33.9220
	(22.3170)	(22.4784)	(25.6568)	(25.8337)
N	2099	2049	2099	2049
Two Day Policy Dummies	N	Y	N	Y

Notes: This table reports the impact on 10-year benchmark G-Sec bond trading volume around the policy announcement days. It is based on equation (2) and uses a daily time series of total G-Sec traded in the secondary market for the period January 2011-August 2019. Robust standard errors of the coefficients are reported in the parentheses. ***- $p < 0.01$, **- $p < 0.05$ and * - $p < 0.1$.

Table 5: No Additional Effect on G-Sec Volatility around RBI Policy Days (after FIT)

	Model			
	(1)	(2)	(3)	(4)
μ_1 : Day before Two-day policy	-1.565***	-1.522***	-1.590***	-1.545***
	(0.393)	(0.398)	(0.393)	(0.398)
μ_2 : First day of two-day policy	0.740	0.684	0.532	0.486
	(0.664)	(0.644)	(0.730)	(0.707)
μ_3 : Second day of two-day policy	0.678	0.617	0.562	0.564
	(0.525)	(0.498)	(0.597)	(0.570)
δ_1 : One day before rate announcement (all policy days)	-0.236	-0.252	-0.031	-0.047
	(0.286)	(0.279)	(0.406)	(0.395)
δ_2 : Day of rate announcement (all policy days)	1.043***	0.727**	0.993**	0.687*
	(0.293)	(0.302)	(0.408)	(0.414)
δ_3 : One day after rate announcement (all policy days)	-0.507***	-0.452***	-0.327**	-0.298**
	(0.109)	(0.125)	(0.132)	(0.143)
δ_4 : Day of surprise policy		0.569***		0.486***
		(0.124)		(0.159)
γ_1 : FIT × One day before rate announcement			-0.417	-0.416
			(0.608)	(0.599)
γ_2 : FIT × Day of rate announcement			0.239	0.376
			(0.648)	(0.691)
γ_3 : FIT × One day after rate announcement			-0.562*	-0.524
			(0.311)	(0.362)
γ_4 : FIT × Day of surprise policy				-0.160
				(0.336)
N	2052	2052	2052	2052

Notes: We estimate the equations (3)-(7) for 10-year benchmark G-Sec yield, using a daily time series from January 2011-August 2019. We report the estimates of the multiplicative heteroskedastic component of the Models 1 to 4 in this table. Specifically, the estimates reported here is based on the following equation:

$$\begin{aligned} s_t = {} & \delta_1 I_t^{MPC-} + \delta_2 I_t^{MPC} + \delta_3 I_t^{MPC+} + \mu_1 I_t^{MP2--} + \mu_2 I_t^{MP2-} + \mu_3 I_t^{MP2} \\ & + \delta_4 I_t^S + I_t^{FIT}\left(\gamma_1 I_t^{MPC-} + \gamma_2 I_t^{MPC} + \gamma_3 I_t^{MPC+} + \gamma_4 I_t^S\right) \end{aligned} \tag{9}$$

The standard errors of the coefficients are reported in the parentheses. ***- $p < 0.01$, **- $p < 0.05$ and * - $p < 0.1$ The Wald test for $\gamma_1 = \gamma_2 = \gamma_3 = 0$ cannot be rejected at $p = 0.1$.

Table 6: No Effect on Stock Market (NIFTY-50 Index) Volatility around RBI Policy Days

	(1)	(2)	(3)	(4)
μ_1 : Day before Two-day policy	-0.318	-0.308	-0.341	-0.341
	(0.501)	(0.503)	(0.504)	(0.505)
μ_2 : First day of two-day policy	-0.144	-0.143	-0.442	-0.443
	(0.820)	(0.825)	(0.845)	(0.846)
μ_3 : Second day of two-day policy	0.779	0.765	0.955	0.956
	(0.646)	(0.652)	(0.674)	(0.674)
δ_1 : One day before rate announcement (all policy days)	0.351*	0.347*	0.643**	0.643**
	(0.188)	(0.188)	(0.269)	(0.269)
δ_2 : Day of rate announcement (all policy days)	-0.156	-0.199	-0.492	-0.489
	(0.291)	(0.300)	(0.437)	(0.458)
δ_3 : One day after rate announcement (all policy days)	-0.427*	-0.421*	-0.251	-0.252
	(0.231)	(0.232)	(0.341)	(0.343)
δ_4 : Day of surprise policy		0.083		-0.003
		(0.115)		(0.152)
γ_1 : FIT $\times$ One day before rate announcement			-0.743*	-0.744*
			(0.433)	(0.434)
γ_2 : FIT $\times$ Day of rate announcement			0.876	0.840
			(0.632)	(0.662)
γ_3 : FIT $\times$ One day after rate announcement			-0.463	-0.456
			(0.465)	(0.471)
γ_4 : FIT $\times$ Day of surprise policy				0.083
				(0.298)
N	2069	2069	2069	2069

Notes: We estimate the equations (3)-(7) for daily returns of NIFTY-50 index, using a time series running from January 2011 to August 2019. We report the estimates of the multiplicative heteroskedastic component of the Models 1 to 4 in this table. Specifically, the estimates reported here is of the following equation:

$$s_t = \delta_1 I_t^{MPC-} + \delta_2 I_t^{MPC} + \delta_3 I_t^{MPC+} + \mu_1 I_t^{MP2--} + \mu_2 I_t^{MP2-} + \mu_3 I_t^{MP2} + \delta_4 I_t^S + I_t^{FIT}\left(\gamma_1 I_t^{MPC-} + \gamma_2 I_t^{MPC} + \gamma_3 I_t^{MPC+} + \gamma_4 I_t^S\right) \quad (10)$$

The standard errors of the coefficients are reported in the parentheses. ***- $p < 0.01$, **- $p < 0.05$ and * - $p < 0.1$ The Wald test for $\gamma_1 = \gamma_2 = \gamma_3 = 0$ cannot be rejected at $p = 0.1$.

Table 7: Effect on Indian Market volatility around FOMC Meeting Days

	G-Sec		Nifty-50	
	(1)	(2)	(3)	(4)
One Day Before Policy	0.0131	0.0209	-0.4504***	-0.4744***
	(0.0974)	(0.0982)	(0.1211)	(0.1234)
Day of Policy	0.2434	-0.0420	0.7342***	0.7595***
	(0.1801)	(0.1979)	(0.1700)	(0.1780)
One Day After Policy	0.0088	0.0004	-0.1158	-0.1346
	(0.1515)	(0.1612)	(0.1468)	(0.1482)
Day of Surprise Policy		0.4680***		0.0409
		(0.0922)		(0.0902)
N	2070	2070	2070	2070

Notes: This table provides the impact on Indian bond (10-year G-Sec) and stock market (NIFTY-50 Index) around the FOMC announcements in the US. The surprise component in Federal Reserve Policy is measured by the change in Fed Funds 1-year Futures rate.

Table 8: Contingency Table for χ^2 test

	BFT statements	FIT statments	Total
Frequency of (growth/inflation) Phrases	a	b	a+b
Frequency of others Phrases	c	d	c+d
Total Phrases	a+c	b+d	N= a+b+c+d

Table 9: Growth and Inflation Related Bi-Grams in Policy Statements (Pre- vs. Post-FIT)

	BFT statements	FIT statements	Total
Panel (a): Growth			
Frequency of Growth Phrases	855	1979	2834
Frequency of others Phrases	2419	5138	7557
Total Phrase	3274	7117	10391
χ^2 Value: 3.2			
Panel (b): Inflation			
Frequency of Inflation Phrases	498	1285	1783
Frequency of others Phrases	2776	5832	8608
Total Phrase	3274	7117	10391
χ^2 Value: 12.8			

Notes: This table provides the χ^2 test based on bi-gram frequency in RBI Policy Statements. The bi-grams are manually classified into growth-related, inflation-related or none. The full list of bi-grams is provided in Appendix Table A4.

Appendix

A FPI Flows explain Impact of FOMC on Indian Markets

The results described above highlight the importance of Fed policy for domestic financial conditions in India. We now show how the channel of foreign portfolio investment (FPI) flows can potentially explain the differential impact of Fed policy on Indian bond and stock markets. We do so by testing for the impact on the volatility of FPI flows using a model similar to the one in Section 5.2. We are interested in FPI flow volatility one day before the FOMC meeting, on the day of the policy announcement and one day after FOMC meeting.[13] If Fed policy affects FPI flows, it should be reflected through a positive value on the coefficient α_2. A positive value implies heightened flows on the day of the FOMC policy announcement. Similarly, there should be less activity on the day before the FOMC meeting, which should be captured through a negative α_1. The results for the above regression are shown in Table A.1.

We find that FPI debt flows growth is lower than average on the day before the Fed policy announcement, but there is no significant change on the day of the policy (see column (1)). Thus, FPI debt flows don't particularly explain the reaction of domestic sovereign bond market in India to FOMC announcements. In the equities market (see column (2)), FPI flows volatility is significantly higher than average on the day of the meeting. Since these results are in line with the results in Table 7 and a lion's share of these FPI flows go into the top-traded firms that constitute the NIFTY-50, FPI equity flows offer a channel to explain the impact of FOMC policy announcements on equity return volatility in India. There are two reasons why these FPI debt flows may not be the channel to influence yield volatility

[13]The press conference announcing the FOMC's decision occurs after the time of market closing in India on the same date. We consider the immediate next day as the one where I_t^{FOMC} take the value of one.

in the Indian sovereign bond market. First, as noted above, the correlation between FPI flows to firms in the NIFTY-50 index and the broader Indian stock markets is likely to be highly positive. But, this is not necessarily the case in the bond markets, where a material portion of the flows may go into corporate bonds and even the flows to the government bond market may be directed at instruments other than the benchmark bond. Second, and more importantly, the bond market has more restrictions placed by the central bank which dampens the impact on FPI flows.

Table A.1: Effect on FPI volume around FOMC Days

	Debt Market	Equity Market
	(1)	(2)
γ_1: One Day Before Policy	-0.0821**	0.0014
	(0.0336)	(0.0104)
γ_2: Day of Policy	0.0676	0.0227*
	(0.0465)	(0.0121)
γ_3: One Day After Policy	-0.0629	0.0207
	(0.0383)	(0.0127)
N	2053	2053

Notes:The following equation is estimated using daily volume of Foreign Portfolio Investment :

$$\Delta Log(Volume_t) = \alpha * \Delta Log(FedFutures_t) + \gamma_1 I_t^{FOMC-} + \gamma_2 I_t^{FOMC} + \gamma_3 I_t^{FOMC+} + Week_t + \eta_t$$

The time series is daily data from January 2011 to August 2019. The standard errors of the coefficients are reported in the parentheses. ***- $p < 0.01$, **- $p < 0.05$ and * - $p < 0.1$

Table A.2: Effect on FPI volume around MPC Days

	Debt Market (1)	Equity Market (2)
μ_1 : Day before Two-day policy	0.0529	-0.0162
	(0.1313)	(0.0212)
μ_2 : First day of two-day policy	-0.0173	-0.0781*
	(0.1438)	(0.0411)
μ_3 : Second day of two-day policy	0.0269	0.0141
	(0.1493)	(0.0316)
δ_1 : One day before policy	0.0289	0.0311
	(0.1189)	(0.0296)
δ_2 : Day of policy	-0.0462	-0.0170
	(0.1103)	(0.0196)
δ_3 : One day after policy	-0.1319**	0.0052
	(0.0652)	(0.0192)
γ_1 : FIT $\times$ One day before policy	-0.0565	-0.0344
	(0.1248)	(0.0330)
γ_2 : FIT $\times$ Day of policy	0.0563	-0.0045
	(0.1223)	(0.0222)
γ_3 : FIT $\times$ One day after policy	0.1116	0.0025
	(0.0788)	(0.0250)
N	2052	2052

Notes:The following equation is estimated using daily volume of Foreign Portfolio Investment :

$$\Delta Log(Volume_t) = \alpha * \Delta Log(OIS_t) + \delta_1 I_t^{MPC-} + \delta_2 I_t^{MPC} + \delta_3 I_t^{MPC+} + \mu_1 I_t^{MP2--} + \mu_2 I_t^{MP2-} + \mu_3 I_t^{MP2} + \mathrm{I}_t^{FIT}\left(\gamma_1 I_t^{MPC-} + \gamma_2 I_t^{MPC} + \gamma_3 I_t^{MPC+}\right) + Week_t + \eta_t$$

The time series is daily data from January 2011 to August 2019. The standard errors of the coefficients are reported in the parentheses. ***- $p < 0.01$, **- $p < 0.05$ and * - $p < 0.1$

Table A.3: Effect on G-Sec volatility around RBI Policy Days (When repo changes)

	(1)	(2)
μ_1 : Day before Two-day policy	-1.498***	-1.520***
	(0.406)	(0.403)
μ_2 : First day of two-day policy	0.678	0.472
	(0.662)	(0.725)
μ_3 : Second day of two-day policy	0.751	0.693
	(0.516)	(0.586)
δ_1 : One day before rate announcement (all policy days)	-0.235	-0.032
	(0.290)	(0.410)
δ_2 : Day of rate announcement (all policy days)	0.907***	0.934**
	(0.292)	(0.408)
δ_3 : One day after rate announcement (all policy days)	-0.520***	-0.434***
	(0.117)	(0.139)
δ_4 : Repo change day	0.193	-0.098
	(0.340)	(0.482)
δ_5 : One day after repo change	-0.026	0.211
	(0.320)	(0.452)
γ_1 : FIT $\times$ One day before rate announcement		-0.421
		(0.615)
γ_2 : FIT $\times$ Day of rate announcement		-0.010
		(0.721)
γ_3 : FIT $\times$ One day after rate announcement		-0.248
		(0.418)
γ_4 : FIT $\times$ Repo change day		0.678
		(0.783)
γ_5 : FIT $\times$ One day after repo change		-0.587
		(0.735)
N	2069	2069

Notes: We estimate the equations (2)-(5) and (7) for 10 year benchmark G-Sec yield, using a daily time series running from January 2011 to August 2019. We report the estimates of the multiplicative heteroskedastic component of the Models 1 to 4 in this table. Specifically, the estimates reported here is of the following equation:

$$s_t = \delta_1 I_t^{MPC-} + \delta_2 I_t^{MPC} + \delta_3 I_t^{MPC+} + \mu_1 I_t^{MP2--} + \mu_2 I_t^{MP2-} + \mu_3 I_t^{MP2} + \delta_4 I_t^{C} + \delta_5 I_t^{C+} + I_t^{FIT}\left(\gamma_1 I_t^{MPC-} + \gamma_2 I_t^{MPC} + \gamma_3 I_t^{MPC+} + \gamma_4 I_t^{C} + \gamma_5 I_t^{C+}\right) \quad \text{(A.11)}$$

The standard errors of the coefficients are reported in the parentheses. ***- $p < 0.01$, **- $p < 0.05$ and * - $p < 0.1$ The Wald test for $\gamma_1 = \gamma_2 = \gamma_3 = 0$ cannot be rejected at $p = 0.1$.

Table A.4: Full list of bi-grams

Inflation Related Phrases			
commodity prices	headline inflation	policy stance	cpi inflation
price index	crude prices	consumer price	global commodity
food inflation	oil prices	excluding food	inflation excluding
inflation expectations	international crude	price situation	crude oil
inflation outlook	retail inflation	food prices	petroleum products
government employees	house rent	pay commission	hra impact
inflation pressures	inflation measured	petroleum gas	inflation trajectory
price pressures	vegetable prices	inflation remained	central pay
liquefied petroleum	inflation path	policy stances	food group
product prices	petroleum product	hra revisions	rent allowances
food items	inflation projections	oil non	baseline inflation
fuel inflation			

Growth Related Phrases			
liquidity conditions	global growth	aggregate demand	current account
credit growth	government securities	capacity utilisation	indian economy
capital flows	external sector	external demand	global economic
services sector	fiscal consolidation	investment demand	monetary developments
export credit	overall outlook	economic conditions	trade deficit
industrial activity	gold imports	export growth	economic activity
industrial production	industrial outlook	gdp growth	investment activity
gva growth	purchasing managers	rural demand	input costs
capital goods	global demand	evolving macroeconomic	macroeconomic situation
global financial	trade tensions	demand conditions	durable liquidity
private investment	direct investment	passenger traffic	value added
manufacturing sector	construction activity	new orders	output gap
private consumption	foreign portfolio	import growth	air passenger
retail sales	wage growth	global trade	services pmi
supporting growth	consumer confidence	manufacturing purchasing	cement production
domestic front	freight traffic	manufacturing activity	core industries
growth remained	economic data	engineering goods	allied activities
gross value	vehicle sales	new business	domestic demand
gold prices	sector activity	order books	consumption expenditure
supply management	net exports	growth prospects	foodgrains production
manufacturing firms	steel consumption	goods production	tourist arrivals
bank credit	domestic economy	rate cuts	world trade
rural wage	capital formation	domestic air	electricity generation
rabi sowing	financing conditions	consumer spending	international prices
public administration	capital market	business sentiment	staff costs
business expectations	consumption demand	gross fixed	commercial vehicle
electronic goods	urban demand	political tensions	railway freight
consumer durables	fixed capital	supply side	foreign tourist
merchandise exports	final consumption	net demand	productive sectors